

Azure Data Engineering Project Roadmap Automated Retail Sales Analytics Pipeline on Azure (Bronze → Silver → Gold)

1. Executive Summary

This project aims to design and implement a **fully automated, scalable data engineering pipeline** using **Microsoft Azure** services. The goal is to process, clean, and visualize **retail sales data** using the **Bronze–Silver–Gold architecture pattern**, ensuring reliable and analytics-ready data for business insights.

By leveraging Azure’s ecosystem — including **Data Factory, Databricks, Synapse Analytics**, and **Power BI** — this solution will automate data ingestion, transformation, and reporting, enabling **real-time decision-making, cost optimization**, and **data governance best practices**.

2. Team members and roles

1. Nagat Osama
 2. Merna Sameh
 3. Ebrahim Attef Abd Elkader
 4. Abdullah Mohamed Hamed
- Each team member contributes to a specific area:
- **Data Engineer** — pipeline design and development.
 - **Cloud Engineer** — Azure infrastructure deployment and automation.
 - **BI Developer** — dashboard design and KPI reporting.
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3. Proposed Solution & Architecture Overview

The project will build an **end-to-end data analytics pipeline** hosted entirely on Azure Cloud.

Data will move through three major layers:

- **Bronze Layer:** Raw data ingestion
- **Silver Layer:** Cleaning and validation
- **Gold Layer:** Aggregation and analytics for reporting

All steps will be automated using Azure services, ensuring scalability and monitoring through Azure DevOps integration.

Architecture Flow

- 1. **Data ingestion** through Azure Data Factory (ADF).
- 2. **Data processing** with Azure Databricks (PySpark).
- 3. **Data storage** in Azure Data Lake Gen2 and Blob Storage.
- 4. **Data warehousing** using Azure Synapse Analytics.
- 5. **Visualization** via Microsoft Power BI dashboards.

4. Resources & Tools

Tool / Service	Purpose / Functionality
Azure Resource Manager	Centralized management and deployment of Azure resources
Azure Servers	Hosting compute workloads and services
Azure Blob Storage	Storing large-scale raw and processed data
Azure Data Factory (ADF)	Orchestration and automation of data pipelines
Azure Databricks	Data processing, transformation, and machine learning (PySpark)
Azure SQL Database	
Azure Synapse Analytics	Data warehousing, analytics, and SQL-based querying
Microsoft Power BI	Interactive data visualization and business intelligence
Azure Key Vault	Secure secrets and credentials management

5. Added Value / Benefits

- Fully **automated** data pipeline from ingestion to reporting
- **Real-time analytics** and dynamic Power BI dashboards
- **Scalable** architecture for large datasets
- **Secure** access via Azure AD and Key Vault
- **Cost-effective** and resource-optimized design
- Supports future **machine learning integration** using Databricks
- Enables **data-driven decision making** through continuous data refreshes

6. Project Roadmap and Phases

Phase 0 — Project Setup & Planning

Goal: Understand architecture, gather sample data, and plan resources.

Deliverables:

- Project plan & architecture diagram
- Azure resource group
- Sample dataset & Git repo

Learning Goals:

- Understand Bronze–Silver–Gold architecture
- Learn project structure and workflow

Key Tasks:

- Create Azure account & resource group
- Draw architecture diagram
- Generate sample data with Python
- Initialize Git repo

Estimated Duration: 3–4 days

Phase 1 — Infrastructure Deployment

Goal: Deploy Azure resources (ADF, Databricks, Data Lake, Key Vault).

Deliverables:

- Azure Data Lake Gen2
- ADF instance
- Databricks workspace
- Key Vault

Learning Goals:

- Learn ARM/Bicep templates
- Understand Azure permissions

Key Tasks:

- Write Bicep template
- Deploy via Azure CLI
- Assign roles & permissions
- Verify deployments

Estimated Duration: 4–6 days

Phase 2 — Bronze Layer: Data Ingestion

Goal: Automate ingestion of raw data into the Bronze layer.

Deliverables:

- ADF Ingest Pipeline
- Databricks ingestion notebook
- Triggered automation

Learning Goals:

- Learn ADF copy activity & triggers
- Understand parameter passing

Key Tasks:

- Build ingestion notebook
- Create ADF pipeline
- Test file ingestion
- Verify bronze data

Estimated Duration: 5–7 days

Phase 3 — Silver Layer: Cleaning & Validation

Goal: Transform and clean data for analytics readiness.

Deliverables:

- Silver transformation notebook
- Data quality checks

Learning Goals:

- Learn PySpark data cleaning
- Apply Delta Lake merges

Key Tasks:

- Write cleaning notebook
- Add validation & deduplication
- Integrate into ADF pipeline

Estimated Duration: 1–2 weeks

Phase 4 — Gold Layer: Aggregation & Analytics

Goal: Produce business-ready metrics and aggregates.

Deliverables:

- Gold aggregation notebook
- Synapse integration

Learning Goals:

- Learn partitioning & Delta optimization
- Learn data modeling for BI

Key Tasks:

- Build aggregation notebook
- Load into Synapse
- Validate aggregates

Estimated Duration: 5–6 days

Phase 5 — Reporting & Visualization

Goal: Build Power BI dashboards from Gold data.

Deliverables:

- Power BI dashboard (.pbix)
- Visual KPIs & filters

Learning Goals:

- Learn Power BI–Azure integration
- Learn DAX basics

Key Tasks:

- Connect Power BI to Synapse or Delta
- Design dashboard visuals
- Schedule refresh

Estimated Duration: 1 week

Phase 6 — Automation, Monitoring & CI/CD

Goal: Add DevOps, alerting, and observability.

Deliverables:

- CI/CD pipelines
- Monitoring & alerts

Learning Goals:

- Learn DevOps for data pipelines
- Add pipeline notifications

Key Tasks:

- Enable ADF Git integration
- Deploy via GitHub Actions / Azure DevOps
- Add alerts & retries

Estimated Duration: 2 weeks

Phase 7 — Optimization & Documentation

Goal: Finalize project, optimize performance, document results.

Deliverables:

- Final README
- Presentation
- Performance metrics

Learning Goals:

- Learn performance tuning best practices
- Create professional documentation

Key Tasks:

- Optimize queries
- Document architecture
- Prepare demo presentation

Estimated Duration: 1 week

1. Timeline Summary

Phase	Description	Estimated Duration
0	Setup & Planning	3–4 days
1	Infrastructure Deployment	4–6 days
2	Data Ingestion (Bronze)	5–7 days
3	Cleaning & Validation (Silver)	1–2 weeks
4	Aggregation & Analytics (Gold)	5–6 days
5	Power BI Reporting	1 week
6	Automation & CI/CD	2 weeks
7	Optimization & Documentation	1 week

2. Technical Specifications

Component	Details
Programming	Python, PySpark, SQL
Data Format	Parquet, Delta Lake
Orchestration	Azure Data Factory
Storage	Azure Data Lake Gen2, Blob Storage
Processing	Azure Databricks
Data Warehouse	Azure Synapse Analytics
Visualization	Microsoft Power BI
Authentication	Azure Active Directory, Managed Identity
Version Control	GitHub / Azure DevOps
Security	Azure Key Vault, Role-Based Access Control (RBAC)

